

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 29

Stationary Distribution

Def: A row vector $\pi = (\pi_i : i \in S)$ of non-negative entries is said to be a stationary (or an invariant) measure for a MC with TPM $P = (p_{ij})_{i,j \in S}$ if

$$\pi_j = \sum_{i \in S} \pi_i p_{ij} \quad (1)$$

for all $j \in S$.

Remark: In matrix notation, (1) can be written as $\pi P = \pi$.

Remark: It is easy to see that for any $n \geq 1$, $\pi P^n = \pi$.

Def: A stationary measure π is called a stationary distribution if $\sum_{i \in S} \pi_i = 1$.

Justification of Terminology

The following theorem justifies the use of the terminology “Stationary”.

Theorem: Let $\{X_n : n \geq 0\}$ be a MC with TPM P and initial distribution $\pi = (\pi_i : i \in S)$. If π is a stationary distribution, then $P(X_n = j) = \pi_j$ for all $n \geq 0$ and for all $j \in S$.

Proof: We will prove by induction. From definition of initial distribution $P(X_0 = j) = \pi_j$. Assume that $P(X_n = j) = \pi_j$. Now, by Markov property,

$$P(X_{n+1} = j) = \sum_{i \in S} P(X_{n+1} = j | X_n = i) P(X_n = i) = \sum_{i \in S} \pi_i p_{ij} = \pi_j$$

for all $j \in S$.

Limiting Distribution (Finite State Space)

The following theorem says that the “limiting” distribution, if it exists, is a stationary distribution.

Theorem: Let S be finite. Suppose that for some $i \in S$,

$$p_{ij}^{(n)} \rightarrow \pi_j \text{ as } n \rightarrow \infty$$

for all $j \in S$. Then $(\pi_j : j \in S)$ is a stationary distribution.

Proof: We have

$$\sum_{j \in S} \pi_j = \sum_{j \in S} \lim_{n \rightarrow \infty} p_{ij}^{(n)} = \lim_{n \rightarrow \infty} \sum_{j \in S} p_{ij}^{(n)} = 1.$$

And

$$\pi_j = \lim_{n \rightarrow \infty} p_{ij}^{(n)} = \lim_{n \rightarrow \infty} \sum_{k \in S} p_{ik}^{(n-1)} p_{kj} = \sum_{k \in S} \lim_{n \rightarrow \infty} p_{ik}^{(n-1)} p_{kj} = \sum_{k \in S} \pi_k p_{kj}.$$

Example

Example 1: Consider the two-state MC with TPM

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{bmatrix} \end{matrix}.$$

- Let (π_0, π_1) be a stationary distribution.
- Then

$$\pi_0 = (1 - \alpha)\pi_0 + \beta\pi_1$$

$$\pi_1 = \alpha\pi_0 + (1 - \beta)\pi_1$$

$$\pi_0 + \pi_1 = 1.$$

- Solving, we get $\pi_0 = \frac{\beta}{\alpha + \beta}$ and $\pi_1 = \frac{\alpha}{\alpha + \beta}$.
- Thus, for this example, stationary distribution exists and is unique.

Example

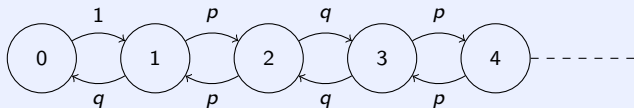
Example 2: Consider a two-state MC with TPM

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

- For any $a \in [0, 1]$, $(a, 1 - a)$ is a stationary distribution.
- Thus, for this example, infinitely many stationary distributions exist.

Example

Example 3: Consider the MC with the following transition diagram ($p + q = 1$, $0 < p < 1$):



- Let $(\pi_i : i \geq 0)$ be a stationary distribution.
- Then

$$\pi_0 = \pi_1 q$$

$$\pi_1 = \pi_0 + \pi_2 p = \pi_1 q + \pi_2 p \implies \pi_1 p = \pi_2 p \implies \pi_1 = \pi_2.$$

- Similarly, $\pi_2 = \pi_3$, $\pi_3 = \pi_4$, $\pi_4 = \pi_5$, ...

Example

- Therefore, $\pi_0 = \pi_1 q$ and $\pi_1 = \pi_2 = \pi_3 = \pi_4 = \dots$
- If $\pi_1 = c$, then $\pi_0 = cq$ and $\pi_1 = \pi_2 = \pi_3 = \dots = c$.
- However, this is not possible as $\sum_{i=0}^{\infty} \pi_i = 1$.
- Thus, in this case, there does not exist a stationary distribution.

Theorem: If $\{X_n : n \geq 0\}$ is a MC with finite state space, then there exist at least one stationary distribution.

Theorems

Theorem: Let $\{X_n : n \geq 0\}$ be an irreducible MC. If π is an invariant measure such that $\pi_i > 0$ for some $i \in S$, then $\pi_k > 0$ for all $k \in S$.

Proof:

- Consider any state $k \neq i$.
- By irreducibility, there exists n with $p_{ik}^{(n)} > 0$. Thus,

$$\pi_k = \sum_{j \in S} \pi_j p_{jk}^{(n)} \geq \pi_i p_{ik}^{(n)} > 0.$$

Theorems

For a fixed state k , consider for each $i \in S$ the expected time spend in i between visits to k :

$$\gamma_i^k = E_k \left(\sum_{n=0}^{T_k-1} \delta_i(X_n) \right).$$

Theorem: Let $\{X_n : n \geq 0\}$ be an irreducible and recurrent MC. Then

- ① $\gamma_k^k = 1$.
- ② $\gamma_j^k = \sum_{i \in S} \gamma_i^k p_{ij}$ for all $j \in S$.
- ③ $0 < \gamma_i^k < \infty$ for all $i \in S$.

Theorems

Theorem: Let $\{X_n : n \geq 0\}$ be an irreducible MC. Fix a state k and let π be an invariant measure with $\pi_k = 1$. Then $\pi \geq \gamma^k$ (entry wise). If in addition the chain is also recurrent, then $\pi = \gamma^k$.

Remark: Let $\{X_n : n \geq 0\}$ be an irreducible and recurrent MC. Let π and μ be two non-zero invariant measures. Then, there exists c such that $\mu = c\pi$.

Proof:

- Fix a state k .
- By the above theorem,

$$\frac{\pi}{\pi_k} = \gamma^k = \frac{\mu}{\mu_k}.$$

- Hence, $\mu = \frac{\mu_k}{\pi_k} \pi = c\pi$, where $c = \frac{\mu_k}{\pi_k}$.